100,000 trillion degrees) that it will explode violently, in a fraction of a second.

The world's dozen experts on the partial marriage of general relativity with quantum theory were quite sure that Hawking had made a mistake. His conclusion violated everything then known about black holes. Perhaps his partial marriage, which differed from other people's, was wrong; or perhaps he had the right marriage, but had made a mistake in his calculations.

For the next several years the experts minutely examined Hawking's version of the partial marriage and their own versions, Hawking's calculations of the waves from black holes and their own calculations. Gradually one expert after another came to agree with Hawking, and in the process they firmed up the partial marriage, producing a new set of physical laws. The new laws are called the *laws of quantum fields in curved spacetime* because they come from a partial marriage in which the black hole is regarded as a non—quantum mechanical, general relativistic, curved spacetime object, while the gravitational waves, electromagnetic waves, and other types of radiation are regarded as *quantum fields*—in other words, as waves that are subject to the laws of quantum mechanics and that therefore behave sometimes like waves and sometimes like particles (see Box 4.1). [A full marriage of general relativity and quantum theory, that is, the fully correct laws of quantum gravity, would treat everything, including the hole's curved space-time, as quantum mechanical, that is, as subject to the uncertainty principle (Box 10.2), to wave/particle duality (Box 4.1), and to vacuum fluctuations (Box 12.4). We shall meet this full marriage and some of its implications in the next chapter.]

How was it possible to reach agreement on the fundamental laws of quantum fields in curved spacetime without any experiments to guide the choice of the laws? How could the experts claim near certainty that Hawking was right without experiments to check their claims? Their near certainty came from the requirement that the laws of quantum fields and the laws of curved spacetime be meshed in a totally consistent way. (If the meshing were not totally consistent, then the laws of physics, when manipulated in one manner, might make one prediction, for example, that black holes never radiate, and when manipulated in another manner, might make a different prediction, for example, that black holes must always